



3D Component Lab

Working interaction, WebGL and AI 3D components for Muse.
Built as an isolated noindex lab, with production changes held
behind verification.

The lab is pushed on `direction/3-the-proof`. Static and browser gates are green. Real GPU rendering remains unverified on this machine.

BRANCH

`direction/3-the-proof`

IMPLEMENTATION TIP

`68cf2a8`

QA MATRIX

80 checks

STATUS

Lab accepted

The lab is working and pushed. No component was promoted into the live ProofWall. The remaining uncertainty is GPU rendering, not build correctness or responsive layout.

■ Passed ■ Unverified or deferred ■ Blocked

BUILD GATES 4 / 4	OBSERVED FAILURES IN COMPLETED CHECKS 0	WEBGL SCENARIOS NOT EXECUTABLE 24 / 80
PASSED FOUR STATIC GATES Production build, TypeScript, scoped ESLint and 15 Meshy tests.	PASSED COMPLETED BROWSER CHECKS EN and AR routes at desktop, mobile and reduced motion. No overflow or noindex failure.	
PASSED PRODUCTION ISOLATION Lenis now runs only inside the noindex demo tree. Production routes use native scrolling.	PASSED MESHY OUTPUT SAFETY A download needs a valid response, size limit and GLB magic bytes before atomic write.	
DEFERRED KAYKIT Needs a legitimate manual itch.io download. Khronos characters remain the test set.	UNVERIFIED GPU RENDERING Twenty-four WebGL route and viewport scenarios need a machine with a working context.	

Recommendation: keep the lab separate. Promote individual components into Direction 4 only after a real GPU visual pass and a per-component product decision.

COMPONENT	CURRENT STATE	PRODUCTION POSITION
Magnetic	Technically tested	Not approved for Direction 4
Crosshair	Technically tested	Not approved for Direction 4
R3F carousel	Working lab demo	Do not replace ProofWall yet
Drei scene	Working lab demo	GPU review required
Characters	Three real Khronos models	Test assets, not final art direction
Meshy	Pipeline tested, API not run	Needs paid key and input rights

02 Lab index

Eight noindex routes hold the working components. The internal banner clears the fixed site header at desktop and mobile sizes.

	<u>Magnetic</u> Hover-pull wrapper around real CTA components (PillButton, Link). ALSO PRODUCTION: HERO CONTACT CTA, CLOSING CTA
	<u>Crosshair cursor</u> SVG reticle that frames hovered interactive elements. Zero WebGL contexts. ALSO PRODUCTION: MOUNTED GLOBALLY IN LAYOUT.TSX
	<u>R3F Carousel</u> Wavy infinite drag/wheel carousel, GLSL wave-distortion shader, Muse's own photos. DEMO-ONLY FOR NOW — PRODUCTION PROOFWALL SWAP IS GATED ON QA (SEE PLAN §3.E)
	<u>ProofWallCarousel</u> The real production candidate for the homepage ProofWall — same content, carousel visual. DEMO-ONLY — HOMEPAGE SWAP DEFERRED, NO GPU AVAILABLE TO QA IT HERE (SEE REPORT)
	<u>Animation core</u> Lenis + gsap ticker, SplitText reveal, Flip layout swap, ScrollTrigger pin. RUNTIME PROOF, NOT A SHIPPABLE EFFECT ON ITS OWN
	<u>drei showcase</u> Glass (MeshTransmissionMaterial), Sparkles, Float, PresentationControls, Environment. DEMO-ONLY
	<u>3D characters</u> Khronos Fox / Cesium Man / Rigged Simple, looping animation, drag-to-orbit. DEMO-ONLY — KAYKIT DEFERRED (ITCH.IO DOWNLOAD NEEDS A MANUAL CLAIM CLICK)
	<u>AI 3D (Meshy)</u> Documented Meshy image-to-3D pipeline; TRELLIS/TripoSr documented, not installed. GENERATION BLOCKED — NO MESHY_API_KEY/PAID PLAN HERE; HONEST EMPTY STATE, NO FAKED ASSET

Magnified source-resolution evidence. The route list is readable.

MEASURED PASS

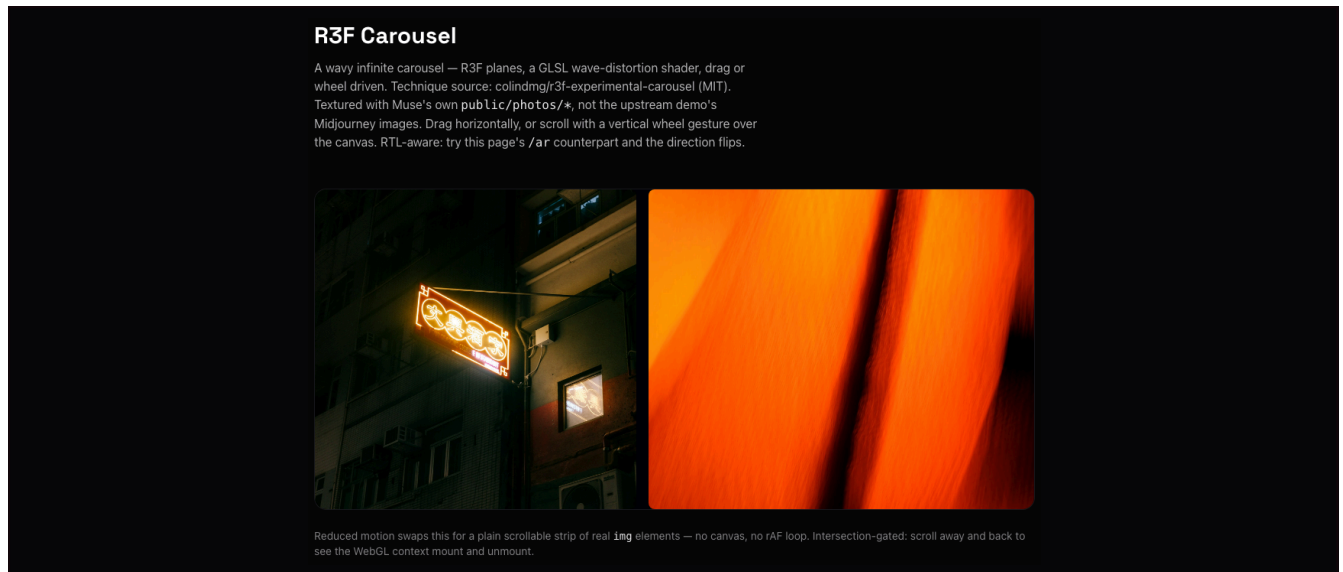
Viewport: 1440 x 900 **HTTP:** 200 **Horizontal overflow:** false **Console errors:** 0 **Robots:** noindex, nofollow.

What is in the lab

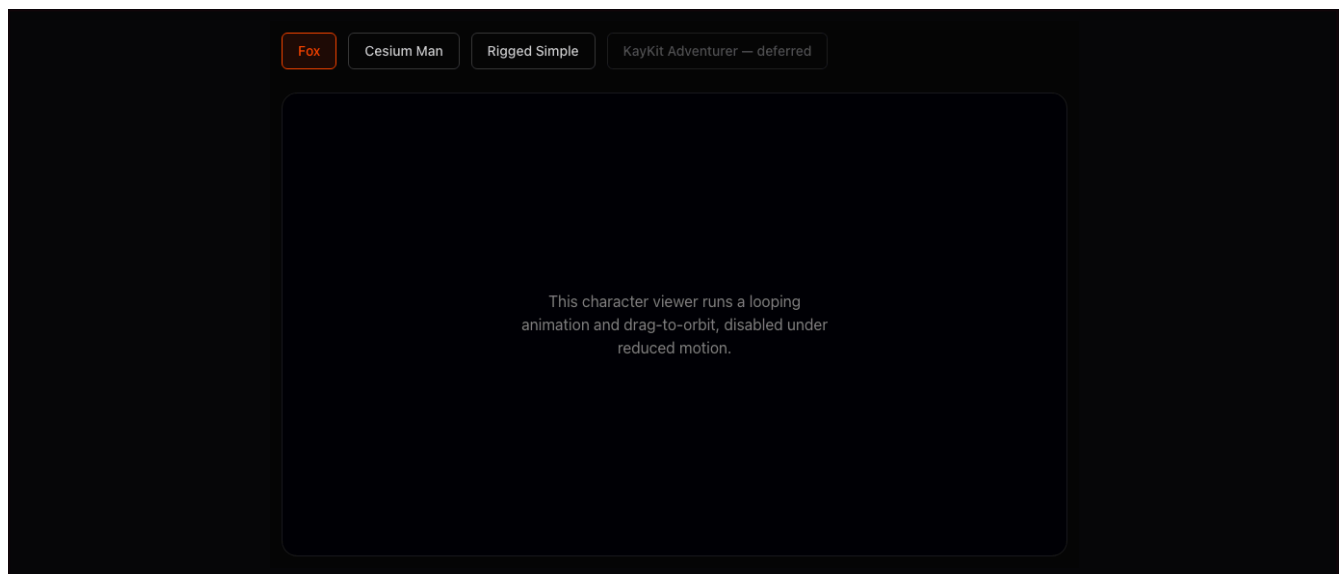
- Magnetic CTA behavior and SVG crosshair interaction
- R3F wave carousel and ProofWall carousel variant
- GSAP animation core and selected drei materials and controls
- Three animated Khronos characters and a reduced-motion fallback
- Meshy AI 3D pipeline with an honest blocked state

03 Accessible fallbacks

Reduced motion does not leave an empty canvas. The carousel becomes real images and the character viewer explains why animation is disabled.



Verified UI text: Reduced motion replaces the animated canvas with a plain scrollable strip of real image elements. There is no canvas or animation-frame loop.



Verified UI text: This character viewer runs a looping animation and drag-to-orbit, disabled under reduced motion.

04 AI 3D pipeline

The Meshy integration is executable and tested, but it has not called the paid API. The page says that directly and shows no fabricated model.

15 TESTS PASSED

Unknown flags, missing values, path traversal, local URLs, invalid responses, invalid GLB data, size limits and atomic writes.

CREDENTIAL HANDLING

The key remains server-side in the CLI environment and is never printed.

PAID API NOT RUN

No key or paid plan was available. No model generation is claimed.

RIGHTS GATE

Any production input image needs confirmed commercial rights before generation.

AI 3D (Meshy)

Plan L3: AI-generated 3D uses the **Meshy API only**. TRELLIS and TripoSR are documented as alternatives below, never installed locally.

GENERATION: BLOCKED / DEFERRED

No model has been generated. There is no MESHY_API_KEY in this environment — that needs a paid Meshy plan (\$20/mo, plan §11 R-6), a real-money decision that hasn't been made, not a missing script. Nothing here is a placeholder or a faked result standing in for a real one.

Magnified 390 px source crop. The complete blocked-state card names the missing key, paid-plan dependency and absence of any placeholder model.

05 Verification and open gates

The code-level gates are green. WebGL checks stay amber because Chromium could not create a context in this environment, including on the pre-existing homepage.

GATE	RESULT	EVIDENCE
Next production build	Passed	56 static pages generated
TypeScript	Passed	<code>tsc --noEmit --incremental false</code>
Scoped ESLint	Passed	All acceptance-phase implementation files
Meshy tests	15 passed	0 failed, 0 skipped
Browser matrix	56 passed	80 total checks, 24 GPU-limited
Genuine browser failures	0	No overflow, console, robots or layout failures outside WebGL context creation

Still open

- **GPU visual pass.** Check carousel distortion, lighting, model animation, context recovery and simultaneous context count.
- **ProofWall decision.** Compare the grid and carousel with LCP, CLS, keyboard and wheel behavior before any swap.
- **KayKit acquisition.** Download through the legitimate itch.io flow, preserve its license and convert only after review.
- **Meshy production run.** Buy a plan only when a Muse-owned input and a concrete use case exist.
- **Direction 4 integration.** Cherry-pick components individually. Do not merge the whole lab branch into the selected site direction.

IMPLEMENTATION BASELINE

The verified implementation ends at 68cf2a8. The PDF and its reproducible source are committed afterward as reporting artifacts.

Final status: accepted as a component lab. Not accepted as proof that every WebGL scene renders correctly on production hardware.